

SUMMER TECHNICAL INTERNSHIP REPORT

*Submitted in partial fulfilment of the requirements for
the award of the degree of*

BACHELOR OF TECHNOLOGY

SUBMITTED BY

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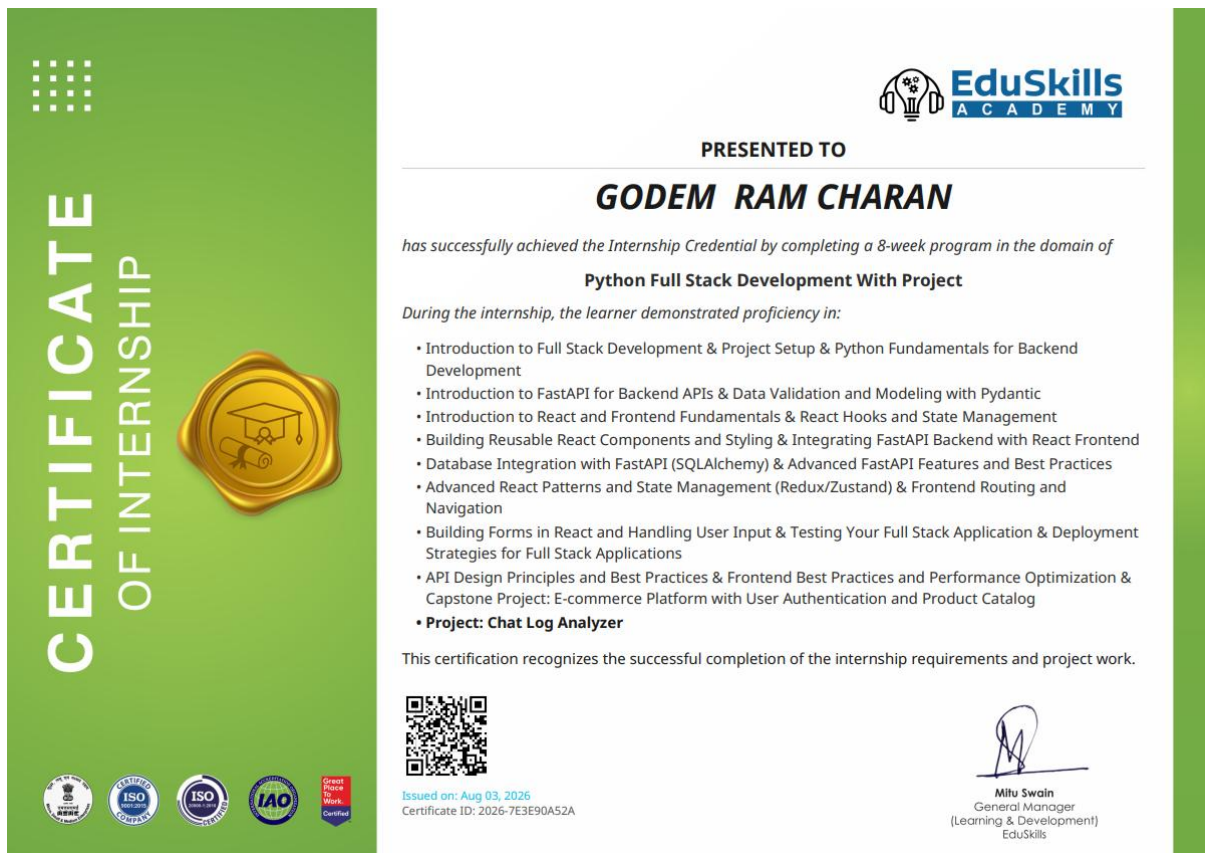
2025–2026

INTERNSHIP CERTIFICATE

(The original certificate of internship completion issued by EduSkills Academy is appended below / attached as a scanned copy along with this report.)

Certificate details —

- Presented to: GODEM RAM CHARAN
- Program: Python Full Stack Development With Project
- Duration: 8-week program (June 8, 2026 – August 3, 2026)
- Issued on: August 3, 2026
- Certificate ID: 2026-7E3E90A52A
- Issuing Authority: Mitu Swain, General Manager (Learning & Development), EduSkills



DECLARATION BY CANDIDATE

I hereby declare that the work presented in this report entitled "Chat Log Analyzer" in the fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering submitted in the Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation, Green Fields, Vaddeswaram is an authentic record of my own work at EduSkills Academy, carried out over a period from June 8, 2026 to August 3, 2026.

The matter embodied in this report has not been submitted for the award of any other degree or diploma.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to EduSkills Academy for providing me the opportunity to undergo this summer technical internship and for the guidance extended throughout the "Python Full Stack Development With Project" program.

I am thankful to my mentors and trainers at EduSkills Academy for their continuous support, valuable feedback, and technical guidance during the course of the internship, which enabled me to successfully complete the capstone project, "Chat Log Analyzer".

I also extend my heartfelt thanks to the Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation, for facilitating this internship opportunity and to my faculty guide for their constant encouragement and support.

Finally, I would like to thank my family and friends for their constant support and motivation throughout this internship.

GODEM RAM CHARAN

ABSTRACT

This report presents an account of the 8-week Summer Technical Internship undertaken at EduSkills Academy under the program "Python Full Stack Development With Project", carried out from June 8, 2026 to August 3, 2026. The internship provided hands-on exposure to full stack web application development using Python-based backend technologies and modern JavaScript frontend frameworks.

During the internship, concepts of backend API development using FastAPI, data validation using Pydantic, database integration using SQLAlchemy, and frontend development using React (including Hooks, state management with Redux/Zustand, routing, and form handling) were studied and applied. The program concluded with the design and development of a capstone project titled "Chat Log Analyzer", which analyzes chat log data and presents insights through a full stack web application with a FastAPI backend and a React-based frontend, backed by a relational database.

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1. Introduction

1.1 Overview of the Host Company

EduSkills Academy is an ed-tech training and certification organization that partners with academic institutions to deliver industry-aligned skilling programs for students across engineering disciplines. It offers structured, mentor-led internship programs that combine self-paced learning modules with hands-on project work, enabling learners to gain practical, job-ready skills in emerging technology domains such as full stack web development, cloud computing, and data science.

The internship undertaken was part of EduSkills Academy's "Python Full Stack Development With Project" track, an 8-week program designed to build proficiency in backend development using Python and FastAPI, and frontend development using React, culminating in a capstone project.

1.2 Objectives of the Internship

- To gain practical exposure to full stack web application development using Python and modern JavaScript frameworks.
- To understand and apply the principles of RESTful API design and implementation using FastAPI.
- To learn data validation and modeling techniques using Pydantic and database integration using SQLAlchemy.
- To build responsive and interactive user interfaces using React, including state management and routing.
- To design, develop, test, and deploy a complete full stack capstone project applying the concepts learned.
- To develop professional skills such as problem-solving, debugging, documentation, and time management.

2. Internship Overview

2.1 Roles and Responsibilities

- Completed self-paced learning modules on Python backend development and React frontend development as per the weekly curriculum.

- Designed and implemented RESTful API endpoints using FastAPI, including request validation with Pydantic models.
- Integrated a relational database using SQLAlchemy ORM for persistent data storage and retrieval.
- Developed reusable, styled React components and implemented client-side routing and state management.
- Connected the React frontend to the FastAPI backend through API integration and handled asynchronous data flows.
- Wrote and executed test cases to validate application functionality and fixed bugs identified during testing.
- Designed and built the capstone project, "Chat Log Analyzer", independently applying the full stack concepts learned during the program.

2.2 Weekly Overview of Tasks

Week	Tasks / Topics Covered
Week 1	Introduction to Full Stack Development & project setup; Python fundamentals for backend development.
Week 2	Introduction to FastAPI for backend APIs; data validation and modeling with Pydantic.
Week 3	Introduction to React and frontend fundamentals; React Hooks and state management.
Week 4	Building reusable React components and styling; integrating FastAPI backend with React frontend.
Week 5	Database integration with FastAPI using SQLAlchemy; advanced FastAPI features and best practices.
Week 6	Advanced React patterns and state management with Redux/Zustand; frontend routing and navigation.
Week 7	Building forms in React and handling user input; testing and deployment strategies for full stack applications.
Week 8	API design principles and best practices; frontend performance optimization; capstone project development — "Chat Log Analyzer".

3. Technical Background

3.1 Software Requirement Specification (SRS)

Functional Requirements:

- The system shall allow chat log files to be uploaded and parsed into structured message records.

- The system shall expose REST API endpoints to retrieve, filter, and analyze chat log data (e.g., message counts, active users, keyword frequency, time-based trends).
- The system shall validate all incoming data using defined schemas before processing or persisting it.
- The system shall persist parsed chat log data and analysis results in a relational database.
- The frontend shall present analyzed data through tables, charts, and summary views, and shall allow users to filter results interactively.

Non-Functional Requirements:

- The API responses should be returned within an acceptable response time for typical chat log file sizes.
- The application should follow modular, maintainable coding practices on both backend and frontend.
- The user interface should be responsive and usable across common screen sizes.
- The application should handle invalid input and errors gracefully with meaningful messages.

3.2 Technologies, Frameworks, and Tools used

Category	Technologies / Tools
Backend	Python, FastAPI, Pydantic
Database / ORM	SQLAlchemy, Relational Database (SQL)
Frontend	React, React Hooks, Redux / Zustand, React Router
API Testing / Docs	FastAPI Swagger UI, Postman
Version Control	Git and GitHub
Development Tools	VS Code, Node.js, npm

4. Project / Task Implementation

4.1 System Architecture and Design

The "Chat Log Analyzer" application follows a three-tier full stack architecture:

- Presentation Layer: A React single-page application that provides the user interface for uploading chat logs, viewing analysis results, and interacting with visualizations and filters.

- **Application Layer:** A FastAPI backend that exposes RESTful endpoints, performs request validation using Pydantic, executes chat log parsing and analysis logic, and communicates with the database layer.
- **Data Layer:** A relational database accessed through SQLAlchemy ORM, used to store uploaded chat logs, parsed messages, and computed analysis results.

The React frontend communicates with the FastAPI backend over HTTP using JSON, following REST conventions. The backend, in turn, interacts with the database through SQLAlchemy models, ensuring a clear separation of concerns between presentation, business logic, and data persistence.

4.2 Module Description and Implementation

Chat Log Upload and Parsing Module:

Accepts chat log files, parses raw text into structured records (sender, timestamp, message content), and validates the parsed data using Pydantic schemas before storing it in the database.

Analysis Engine Module:

Processes stored chat data to compute statistics such as total message counts, per-user message counts, most active time periods, and frequently used keywords or phrases.

API Module:

Exposes FastAPI endpoints for uploading logs, triggering analysis, and retrieving results, with appropriate request/response models and error handling.

Frontend Dashboard Module:

A React-based dashboard that lets users upload chat logs, view summary statistics, filter results by user or date range, and visualize trends, built using reusable components and managed application state.

Database Module:

SQLAlchemy models representing chat sessions, messages, and analysis results, with relationships defined to support efficient querying.

4.3 Application / System Screenshots

[Insert screenshots of the application here: home/upload screen, analysis dashboard, and API documentation (Swagger UI) screen.]

5. Conclusion and Future Scope

5.1 Summary of Skills and Learning Outcomes

- Gained practical proficiency in building and consuming RESTful APIs using FastAPI and Pydantic.
- Learned to design and integrate relational database models using SQLAlchemy ORM.
- Developed skills in building modern, component-based frontend applications using React, including hooks, state management, and routing.
- Understood the end-to-end workflow of full stack application development, from API design to frontend integration, testing, and deployment.
- Improved problem-solving, debugging, and independent project execution skills through the completion of the capstone project.

5.2 Future Enhancements

- Add support for analyzing chat logs from multiple platforms and file formats.
- Incorporate natural language processing techniques for sentiment analysis and topic detection.
- Add user authentication and role-based access control for multi-user usage.
- Enhance the dashboard with richer, interactive data visualizations.
- Optimize backend performance for analyzing larger chat log datasets.

6. References

- FastAPI Documentation — <https://fastapi.tiangolo.com/>
- React Documentation — <https://react.dev/>
- SQLAlchemy Documentation — <https://docs.sqlalchemy.org/>
- Pydantic Documentation — <https://docs.pydantic.dev/>
- EduSkills Academy — Python Full Stack Development With Project, Internship Program materials.